Topic: Rank-Nullity Theorem

Question 1.

(a) Let $\boldsymbol{S} : \mathcal{P}_5(\mathbb{R}) \to \mathcal{M}_{2\times 2}(\mathbb{R})$ be a surjective linear map. What is $\mathrm{nullity}(\boldsymbol{S})$?

(b) Let $\boldsymbol{T} : \mathcal{M}_{3\times 2}(\mathbb{R}) \to \mathcal{P}_5(\mathbb{R})$ be an injective linear map. What is $\mathrm{rank}(\boldsymbol{T})$?

Question 2. Let $\boldsymbol{T}\colon \mathbb{R}^3 \to \mathbb{R}$ be linear. Show that there exist real numbers a, b and c such that

$$T((x, y, z)^T) = ax + by + cz, \text{ for all } (x, y, z) \in \mathbb{R}^3.$$

Describe geometrically the possibilities for the null space of T.

$|B| = \dim(V)$

Question 3. Consider the *linear* map $\boldsymbol{T} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ defined by

* range ≠ codomain

everything you can make

vectorspace

(A) is hard to prove

$$\boldsymbol{T}(p(x)) = xp(x) + p'(x).$$

(a) Find a basis for $\mathrm{range}(\boldsymbol{T})$. Is $\boldsymbol{T}$ surjective?

$p(x) = a + bx + cx^2$

(b) Is $\boldsymbol{T}$ injective?

rank = dim (basis of range(T))

nullity = dim (basis of null(T))

* we rank nullity theorem if don't know the map

Question 4.

(a) Suppose that $\boldsymbol{T} : \mathcal{V} \to \mathcal{W}$ is a linear map between vector spaces $\mathcal{V}$ and $\mathcal{W}$.

Prove that $\boldsymbol{T}$ is injective (one-to-one) if and only if $\boldsymbol{T}$ maps linearly independent subsets of $\mathcal{V}$ to linearly independent subsets of $\mathcal{W}$.

(b) Suppose $\boldsymbol{T} : \mathcal{V} \to \mathcal{W}$ is an injective linear map and $\mathcal{S}$ is a subset of $\mathcal{V}$. Prove that

$\mathcal{S}$ is linearly independent if and only if $\boldsymbol{T}(\mathcal{S})$ is linearly independent.

→ injective, surjective ⇒ dim(W) = dim(V) = k

Question 5. Let $T : \mathcal{V} \to \mathcal{W}$ be an isomorphism, where $\mathcal{V}$ and $\mathcal{W}$ are finite-dimensional vector spaces. Prove that $\mathcal{B} = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_k\}$ is a basis for $\mathcal{V}$ if and only if $\mathcal{A} = \{\boldsymbol{T}(\boldsymbol{v}_1), \ldots, \boldsymbol{T}(\boldsymbol{v}_k)\}$ is a basis for $\boldsymbol{W}$.

Question 4 is very useful here.

Condition (B)
span(A) = W
↳ see if linear combi of A can make everything in W.

$|A| = k$

Question 6. Let $\boldsymbol{B} \in \mathcal{M}_{n\times n}(\mathbb{F})$ be an invertible matrix. Define $\Phi : \mathcal{M}_{n\times n}(\mathbb{F}) \to \mathcal{M}_{n\times n}(\mathbb{F})$ by

$$\Phi(\boldsymbol{A}) = \boldsymbol{B}^{-1}\boldsymbol{A}\boldsymbol{B}.$$

Prove that Φ is an isomorphism.

Question 7. Let $\mathcal{V}$ and $\mathcal{W}$ be vector spaces and $\boldsymbol{T}, \boldsymbol{U} : \mathcal{V} \to \mathcal{W}$ be nonzero linear maps. Prove that if $\mathrm{range}(\boldsymbol{T}) \cap \mathrm{range}(\boldsymbol{U}) = \{\boldsymbol{0}\}$, then $\{\boldsymbol{T}, \boldsymbol{U}\}$ is a linearly independent subset of $\mathcal{L}(\mathcal{V}, \mathcal{W})$.

Topic: Matrix Representation of Linear Maps

Question 8. Define $\boldsymbol{T} : \mathcal{M}_{2\times 2}(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ by

$$\boldsymbol{T}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = (a+b) + 2dx + bx^2.$$

Let

$$\mathcal{B} = \left\{ \boldsymbol{E}_{1,1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \boldsymbol{E}_{1,2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \boldsymbol{E}_{2,1} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \boldsymbol{E}_{2,2} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\} \text{ and } \mathcal{A} = \{1, x, x^2\}$$

(a) Determine $[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{B}}$.

(b) Compute $[\boldsymbol{T}]^{\mathcal{C}}_{\mathcal{B}}$, where $\mathcal{C} = \{x^2, x^2+1, x^2+x\}$.

(c) Using the matrix representations $[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{B}}$ or $[\boldsymbol{T}]^{\mathcal{C}}_{\mathcal{B}}$, find a basis for $\operatorname{null}(\boldsymbol{T})$.

(d) Find a basis for $\operatorname{range}(\boldsymbol{T})$.

Question 9. Let $\mathcal{A}$ denote the standard basis of $\mathbb{R}^2$, let $\mathcal{B}$ denote the standard basis of $\mathbb{R}^3$, and let $\mathcal{C}$ denote the following basis of $\mathbb{R}^3$:

$$\mathcal{C} = \Big\{(1,1,1), (1,-1,0), (1,0,-1)\Big\}.$$

Consider the matrix-multiplication map $\boldsymbol{T} : \mathbb{R}^3 \to \mathbb{R}^2$ with the matrix $\boldsymbol{A} = \begin{bmatrix} 1 & 0 & 0.1 \\ -2 & 2.5 & 0 \end{bmatrix}$.

(a) What is $[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{B}}$?

(b) Determine $[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{C}}$.

Question 10. Let $\boldsymbol{T} : \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ and $\boldsymbol{U} : \mathcal{P}_2(\mathbb{R}) \to \mathbb{R}^3$ be the linear maps respectively defined by

$$\boldsymbol{T}(f(x)) = (3+x)f'(x) + 2f(x) \text{ and } \boldsymbol{U}(a+bx+cx^2) = (a+b, c, a-b).$$

Let $\mathcal{A}$ and $\mathcal{B}$ be the standard ordered bases of $\mathcal{P}_2(\mathbb{R})$ and $\mathbb{R}^3$, respectively.

(a) Compute $[\boldsymbol{U}]^{\mathcal{B}}_{\mathcal{A}}$, $[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{A}}$, and $[\boldsymbol{UT}]^{\mathcal{B}}_{\mathcal{A}}$ directly. Verify that $[\boldsymbol{UT}]^{\mathcal{B}}_{\mathcal{A}} = [\boldsymbol{U}]^{\mathcal{B}}_{\mathcal{A}}[\boldsymbol{T}]^{\mathcal{A}}_{\mathcal{A}}$.

(b) Let $h(x) = 3 - 2x + x^2$. Verify that $[\boldsymbol{U}(h(x))]_{\mathcal{B}} = [\boldsymbol{U}]^{\mathcal{B}}_{\mathcal{A}}[h(x)]_{\mathcal{A}}$.

Question 11. Consider the vector space $\mathbb{C}$ over $\mathbb{R}$ (that is $\mathcal{V} = \mathbb{C}$ and $\mathbb{F} = \mathbb{R}$), the map $\boldsymbol{T} : \mathbb{C} \to \mathbb{C}$ which maps a complex number z to its complex conjugate $\bar{z}$ (that is, it takes the complex number $a + bi$ to $a - bi$, where $a, b \in \mathbb{R}$).

(a) Prove that the map $\boldsymbol{T}$ is linear.

(b) Compute $[\boldsymbol{T}]^{\mathcal{B}}_{\mathcal{B}}$, where $\mathcal{B}$ is the basis $\mathcal{B} = \{1, i\}$ of $\mathbb{C}$.